

Exact local obstructions around a 64-modular Hadamard matrix of order 668

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Abstract

Eliahou recently constructed a 64-modular Hadamard matrix of order 668 from a structured length-167 Golay quadruple modulo 64. We study two sharply delimited exact-repair questions around that construction. First, if Eliahou’s sequence $q = (+^{83}, -^2, +^{81}, -)$ is held fixed, then no choice of the other binary sequence makes the special quadruple exactly Golay. The proof is a parity telescope followed by even–odd decimation: any such repair would produce a Turyn sequence in $TU(41)$, which is empty. We independently reproduce that endpoint by an exact 461-shard enumeration. Second, allowing both sequences to vary is equivalent to the base-sequence problem $BS(84, 83)$. A finite decomposition by ordinary and alternating margins, endpoint-quad parity, and primitive-root norm identities reports no exact base sequence within raw labelled Hamming distance 18 of the published seed. The latter is a local computational claim about one orientation of one structured construction, not a nonexistence result for $BS(84, 83)$ or for Hadamard matrices of order 668. Exact data, bounded-memory generators, and independent artifact checkers are being assembled for release with the paper; proof-grade certification of all solver-infeasible leaves is a release requirement.

Verification disclaimer. The human author is a complete amateur exploring the limits of AI-assisted mathematics and cannot independently validate these claims. This is an unreviewed artifact for expert checking, not an established result. Passing the included programs establishes properties of their encoded finite models; it is not peer review and does not by itself establish priority.

1 Introduction

A Hadamard matrix of order n is a matrix $H \in \{\pm 1\}^{n \times n}$ satisfying

$$HH^\top = nI_n.$$

The Hadamard conjecture asserts existence whenever $4 \mid n$. The smallest order for which no matrix is presently known is 668; the next unresolved orders in the implemented range are 716, 892, and 1132 [1, 9].

Eliahou constructed a 64-modular matrix at order 668, meaning

$$HH^\top \equiv 668I_{668} \pmod{64},$$

from four binary sequences of length 167 and the Goethals–Seidel array [6]. A subsequent paper embeds this example in families of 64-modular constructions [7]. The order-668 seed is unusually close to exact complementarity: only thirteen positive aperiodic correlation lags are nonzero.

The natural question is whether a small sign repair can turn this structured seed into an exact Golay quadruple and hence into a true Hadamard matrix. The present paper proves one bounded negative result and reports one larger solver-backed exclusion.

Theorem 1 (Fixed- q obstruction). *Keep Eliahou's length-167 sequence*

$$q = (+^{83}, -^2, +^{81}, -)$$

and the special form

$$(s, s', sq, (sq)').$$

No binary sequence s makes this quadruple exactly Golay.

Computational Claim 2 (Raw radius-18 solver report). *Let $\mathcal{E} \in \{\pm 1\}^{334}$ be the labelled base quadruple obtained from Eliahou's published order-668 seed. The recorded CP-SAT decomposition reports that there is no*

$$(A; B; C; D) \in \text{BS}(84, 83)$$

with raw labelled Hamming distance

$$\text{dist}((A; B; C; D), \mathcal{E}) \leq 18.$$

The word *raw* is essential. No distance-changing equivalence is applied, and no global neighbourhood of an equivalence class is claimed. The ball has radius 18 in a 334-dimensional binary cube; even a certified exclusion would not suggest that the unrestricted base-sequence problem is nearly exhausted.

Theorem 1 is a short mathematical reduction whose endpoint uses the exhaustive classification of Turyn sequences of long length below 43 [5]. Claim 2 is a finite solver report. Its decomposition and decoded root witnesses have dependency-free checkers. The public release is not proof-grade until every solver-reported infeasible leaf is also covered by an independently replayable proof or a second exact enumeration; Section 5 makes that distinction explicit.

2 Sequences, the modular seed, and base sequences

For a sign sequence $X = (x_0, \dots, x_{r-1})$, write

$$N_X(k) = \sum_{i=0}^{r-k-1} x_i x_{i+k} \quad (0 \leq k < r)$$

for its aperiodic autocorrelation. A quadruple (X_1, X_2, X_3, X_4) of equal length is Golay when

$$\sum_{j=1}^4 N_{X_j}(k) = 0 \quad (1 \leq k < r).$$

Let $h \in \{\pm 1\}^{167}$ be $+1$ on coordinates $0, \dots, 83$ and -1 on coordinates $84, \dots, 166$, and write $s' = hs$ for pointwise multiplication. Eliahou's quadruple is

$$\mathcal{G}(s, q) = (s, hs, sq, hsq). \tag{1}$$

The published q has alternating run lengths

$$(83, 2, 81, 1).$$

For reproducibility, the published s has run-length word

$$((4)^5, (2, 1, 1)^5, 1, 5, (4)^4, (2, 1, 1)^6, (4)^4, 3, \\ (1, 2, 1)^5, 3, (4)^4, 3, (1, 2, 1)^5),$$

where an exponent means repetition of a block and the first run has sign $+1$. These two words are encoded directly in run-length form in the accompanying checker.

The nonzero summed aperiodic correlations of the quadruple occur at the thirteen lags shown below.

k	4	8	12	16	26	30	34	38	42	46	50	54	58
$\sum_j N_{X_j}(k)$	-512	384	-256	128	-64	128	-192	256	-320	256	-192	128	-64

Thus every residual is divisible by 64, but the quadruple is not exact.

A base sequence in $\text{BS}(m, n)$ is a quadruple $(A; B; C; D)$ with $|A| = |B| = m$, $|C| = |D| = n$, and

$$N_A(k) + N_B(k) + N_C(k) + N_D(k) = 0 \quad (1 \leq k < m),$$

where the shorter terms vanish once $k \geq n$.

Proposition 3 (Exact change of variables). *The special quadruple $\mathcal{G}(s, q)$ of length 167 is Golay if and only if*

$$A = s_{[0,84)}, \quad B = (sq)_{[0,84)}, \quad C = s_{[84,167)}, \quad D = (sq)_{[84,167)}$$

belong to $\text{BS}(84, 83)$. The correspondence is bijective, with inverse

$$s = A \| C, \quad q = (AB) \| (CD).$$

Proof. At lag k , the coefficient in the summed autocorrelation of (1) is

$$\sum_i s_i s_{i+k} (1 + h_i h_{i+k}) (1 + q_i q_{i+k}).$$

Cross-half terms vanish. The remaining expression is twice

$$N_A(k) + N_B(k) + N_C(k) + N_D(k).$$

It vanishes automatically for $k \geq 84$, proving the equivalence. The displayed inverse follows coordinatewise. $\square$

Proposition 3 is useful in both directions: an exact repair gives a standard base sequence, and any base-sequence candidate gives explicit (s, q) and a full order-668 Goethals–Seidel verification path.

3 The fixed- q obstruction

We prove Theorem 1. The special shape of the four runs in q makes almost all variables disappear from the correlation equations. Write

$$X = s_{[0,83)}, \quad u = s_{84}, \quad Y = s_{[85,166)}, \quad v = s_{166}.$$

The coordinate s_{83} is isolated. Direct inspection of the factor $(1 + h_i h_{i+k})(1 + q_i q_{i+k})$ in (1) leaves the within- X and within- Y pairs at lags $1 \leq k \leq 80$, only the two within- X pairs at lag 81, and the

pair x_0x_{82} together with uv at lag 82. No active pair remains at a larger lag. Exact complementarity is therefore equivalent to

$$N_X(k) + N_Y(k) = 0 \quad (1 \leq k \leq 80), \quad (2)$$

$$x_0x_{81} + x_1x_{82} = 0, \quad (3)$$

$$x_0x_{82} + uv = 0. \quad (4)$$

Lemma 4 (Parity telescope). *Every solution of (2)–(4) satisfies*

$$x_{82-i} = (-1)^{i+1}x_i \quad (0 \leq i \leq 82), \quad (5)$$

$$y_{80-i} = (-1)^iy_i \quad (0 \leq i \leq 80), \quad (6)$$

$$u = v. \quad (7)$$

Proof. For $0 \leq k \leq 81$, let P_k be the product of all sign terms in $N_X(k) + N_Y(k)$, with the Y product empty at $k = 81$. Every lag-zero term is a square, so $P_0 = 1$. At positive lag the zero sum contains $2(82 - k)$ signs, of which exactly $82 - k$ are negative, so

$$P_k = (-1)^k.$$

Cancellation between consecutive products gives

$$(x_ix_{82-i})(y_iy_{80-i}) = -1 \quad (0 \leq i \leq 80).$$

Put $r_i = x_ix_{82-i}$ and $t_i = y_iy_{80-i}$. Reflection and comparison at i and $80 - i$ give $r_i = r_{i+2}$. The central products are $r_{41} = t_{40} = 1$, while $r_{40}t_{40} = -1$ gives $r_{40} = -1$. These two central values fix the odd and even classes of r_i ; the relation $r_it_i = -1$ then fixes t_i . This gives (5) and (6). In particular $x_0x_{82} = -1$, so equation (4) gives $uv = 1$ and hence $u = v$. $\square$

Separate the even and odd coordinates:

$$\begin{array}{lll} E = (x_0, x_2, \dots, x_{82}), & |E| = 42, & E^{\text{rev}} = -E, \\ O = (x_1, x_3, \dots, x_{81}), & |O| = 41, & O^{\text{rev}} = O, \\ P = (y_0, y_2, \dots, y_{80}), & |P| = 41, & P^{\text{rev}} = P, \\ Q = (y_1, y_3, \dots, y_{79}), & |Q| = 40, & Q^{\text{rev}} = -Q, \end{array}$$

and set $B = (u, Q, v)$. Since Q is skew and $u = v$, the endpoint contributions to $N_B(k)$ cancel for $1 \leq k \leq 40$. The even-lag equations in (2), together with (4), become

$$N_E(k) + N_B(k) + N_O(k) + N_P(k) = 0 \quad (1 \leq k \leq 41).$$

Therefore

$$(E; B; O; P) \in \text{BS}(42, 41).$$

Its first long sequence is skew and its first short sequence is symmetric, which is precisely a Turyn sequence in TU(41) under the standard definition [3]. Edmondson, Seberry, and Anderson exhaustively enumerated all inequivalent Turyn sequences whose long length is below 43 and found none of long length 42 [5]. This proves Theorem 1. As an independent modern reproduction of that endpoint, the accompanying outside-in enumerator exhausts all 461 normalized depth-five shards for TU(41): 57,543,021 deterministic nodes and zero solutions. A separate Python program exhausts the 2^{19} assignments defining the shard cover.

A necessary correction. The argument does not follow from 83 failing to be a sum of two squares. For BS(42, 41) the relevant row-sum identity at odd parameter 41 is

$$C^2 + D^2 = 4 \cdot 41 - 2 = 162 = 9^2 + 9^2.$$

The final nonexistence step is the Turyn classification, not a two-squares obstruction.

4 The raw seed neighbourhood

Let

$$\mathcal{E} = (A_0; B_0; C_0; D_0)$$

be the labelled base quadruple obtained from the published (s, q) through Proposition 3. We now allow all 334 signs to change.

4.1 Margins and endpoint quads

For a sign sequence X , let

$$S(X) = \sum_i x_i, \quad T(X) = \sum_i (-1)^i x_i.$$

Evaluation of the base-sequence norm identity at $z = 1$ gives

$$S(A)^2 + S(B)^2 + S(C)^2 + S(D)^2 = 334,$$

and evaluation at $z = -1$ gives the same identity for T . The long sums are even, the short sums odd. Independent sequence negations and equal-length swaps reduce the ordinary sums to twelve canonical profiles:

$$\begin{array}{cccc} (4, 2, 17, 5) & (6, 0, 17, 3) & (8, 6, 15, 3) & (10, 0, 15, 3) \\ (10, 4, 13, 7) & (10, 8, 11, 7) & (10, 8, 13, 1) & (12, 10, 9, 3) \\ (14, 4, 11, 1) & (14, 8, 7, 5) & (16, 2, 7, 5) & (18, 0, 3, 1). \end{array}$$

Coordinate parity leaves exactly twenty-four compatible alternating profiles for each ordinary profile, hence 288 nominal margin shards. The verifier expands every raw labelled image under the normalizing negations, swaps, and safe reversals before measuring distance from $\mathcal{E}$.

For $1 \leq k \leq 82$, the lag- k equation contains $2(167 - 2k)$ sign terms, while the lag-83 equation contains two. A zero sum forces exactly half to be negative; that half-count is odd, so the term product is -1 at every positive lag. The lag-zero product is $+1$. Dividing consecutive products telescopes to the standard base-sequence endpoint parity system, equivalently the endpoint-quad conditions

$$\begin{aligned} A_j A_{83-j} B_j B_{83-j} &= \begin{cases} -1, & j = 0, \\ +1, & 1 \leq j \leq 41, \end{cases} \\ C_j C_{82-j} D_j D_{82-j} &= +1 \quad (0 \leq j \leq 40). \end{aligned}$$

The two central short signs C_{41} and D_{41} are unpaired and remain unrestricted by these conditions. The modular seed already has these products. Thus a repair must flip an even number of signs in each four-cell endpoint quad.

For one sequence, (S, T) fixes the even- and odd-coordinate sums:

$$E_X = \frac{S+T}{2}, \quad O_X = \frac{S-T}{2}.$$

The minimum number of flips required for a target pair is consequently exact. A dynamic program processes the endpoint quads independently, retaining the cheapest realization of every eight-component margin delta.

Computational Lemma 5 (Dependency-free inner ball). *No exact base sequence lies within raw distance 13 of $\mathcal{E}$. At radius 13 there are 85 raw margin targets and none survives the endpoint-quad dynamic program. At exact distance 14, eighteen targets survive.*

The unique closest raw margin target occurs at distance 8:

$$A : (-18, 18), \quad B : (0, 0), \quad C : (3, 1), \quad D : (-1, -3).$$

It requires eight positive odd coordinates of A to flip. They lie in eight distinct endpoint quads, so the correct quad products cannot be preserved at distance 8. The dynamic program generalizes this observation through radius 13 and is independently checked against complete brute force on small even- and odd-length fixtures.

4.2 Primitive-root filters

Write $X(z) = \sum_i x_i z^i$. For $m \in \{3, 4, 6\}$, evaluation at a primitive m th root expresses the norm of each sequence by two integer linear forms $a_m(X), b_m(X)$:

$$|X(\zeta_m)|^2 = a_m(X)^2 + \epsilon_m a_m(X) b_m(X) + b_m(X)^2, \quad (\epsilon_3, \epsilon_4, \epsilon_6) = (-1, 0, 1).$$

Every base sequence satisfies

$$\sum_{X \in \{A, B, C, D\}} |X(\zeta_m)|^2 = 334 \quad (m = 3, 4, 6). \quad (8)$$

The finite models encode the two linear forms directly and impose their quadratic norm through a complete allowed-value table. Terms above 334 can be discarded because all four norms are nonnegative.

The remaining frontier is split by raw margin target. Each model fixes all eight ordinary and alternating class sums, the exact distance interval, every endpoint-quad product, and (8). For the hard distance-18 root cases, endpoint quads are grouped into exact orbits by their residues modulo 12 and their oriented seed signs. Any quotient witness is decoded to all 334 physical signs and rechecked with integer arithmetic.

The completed counts are:

scope	targets	root-infeasible	root survivors
complete ball through radius 16	197	197	0
exact distance-17 shell	276	276	0
exact distance-18 shell	823	811	12

Solver models with ordinary length-7 compression report nine of the twelve distance-18 root-survivor targets infeasible. Applying the same compression after coordinate alternation, equivalently primitive-14 compression, reports the final three infeasible. This supports Claim 2 subject to the proof-certification boundary made explicit next.

5 Certificates and trust boundary

The reproducibility package separates four levels of evidence.

1. Standard-library programs reconstruct all raw margin targets and prove the radius-13 margin-plus-quad exclusion by dynamic programming.
2. Independent artifact checkers reconstruct the 197, 276, and 823 frontiers, verify target uniqueness and completeness, pin all file hashes, and validate every artifact-selection edge.
3. All twelve decoded root witnesses are independently checked against the margin, distance, endpoint-quad, and primitive-3/4/6 identities.
4. The 1,284 root-infeasible leaves ($197 + 276 + 811$), together with the twelve primitive-7/14 compression-infeasible leaves, were originally reported by one-worker OR-Tools CP-SAT models. Their JSON records are reproducible solver outcomes, not independently replayable UNSAT proofs.

Four representative leaves—one each from radius 16, shell 17, shell 18, and shell-18 primitive-7 compression—have now been translated independently to deterministic CNF. Their pinned binary DRAT traces regenerate and pass `drat-trim`; replay of all four took 34.62 seconds and peaked at 250 MB resident memory. A known feasible shell-18 root instance also passes an independent positive-model checker with the symmetry quotient enabled. All twelve stored root witnesses separately extend to SAT models of their exactly pinned, unquotiented v2 CNFs and pass every clause plus independent arithmetic checks. An exhaustive contribution-signature regression independently verifies the global root orbit partition for every even endpoint-quad mask. This is 4/1,296 proof coverage, not certification of the complete claim.

The release gate for a theorem-level preprint is to replace all 1,296 infeasible solver outcomes in item 4 by proof-producing SAT or pseudo-Boolean certificates, or by a structurally independent exact enumeration of every leaf. Until that gate is met, the radius-18 statement should be read as a strongly audited computational claim rather than a proof-assistant-level theorem. The fixed- q reduction is independent of these solver records.

All production runs were single-threaded. The original frontier solves peaked at 176 MB resident memory; the dependency-free shell-18 artifact checker peaked at 39 MB. The independent TU(41) enumeration completed 461/461 shards in 362.92 seconds of summed serial time; its slowest shard used 1.38 MB peak resident memory. Commands, solver versions, wall times, memory measurements, and SHA-256 digests are preserved in the repository.

6 Scope and relation to other work

The conclusions are deliberately local.

- Theorem 1 fixes the exact published q . It does not exclude changing both s and q , a nonspecial Golay quadruple, a Legendre pair, or a general Hadamard matrix.
- Claim 2 concerns one raw labelled ball. It does not exclude the rest of BS(84, 83).
- A Legendre pair of length 333 would independently construct an order-668 Hadamard matrix. Current compression-based work on that route is mathematically distinct from the seed-repair problem here [8, 2].

Doković classified $BS(n+1, n)$ through $n = 30$ [4]; that enumeration does not reach the present $n = 83$. At the audit cutoff, no public source located by the authors stated either the fixed- q reduction to TU(41) or the radius-18 repair exclusion. That is provisional search evidence, not a guarantee of worldwide priority. Project policy prohibits external outreach, so no unpublished-overlap claim is made.

Code and data availability

The development repository is https://github.com/AlecKriebel/Math/tree/main/hadamard_668_search. It contains the published seed reconstruction, finite-model generators, pinned solver records, and independent artifact checkers. The present manuscript and certificate work are not yet an immutable public release. The current reproduction environment uses Python 3.12.13 with OR-Tools pinned at 9.14.6206; this PDF is built with Tectonic 0.16.9. Before public circulation they will be frozen at a tagged commit with a complete manifest and, if size requires, separately archived proof files identified by SHA-256 digests.

AI assistance and responsibility

The derivations, programs, certificates, literature audit, and manuscript were developed with heavy assistance from ChatGPT 5.6 Sol under Alec Kriebel’s direction. Alec Kriebel is the named human author and has requested prominent disclosure that he is not qualified to validate the mathematics independently. Independent expert scrutiny is essential.

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